unacademy

COA 2.0

Main Memory: RAM & ROM

By: Vishvadeep Gothi

→ Memory

→ Representatⁿ: no. of cells × 1 cell capacity

Are You Ready?

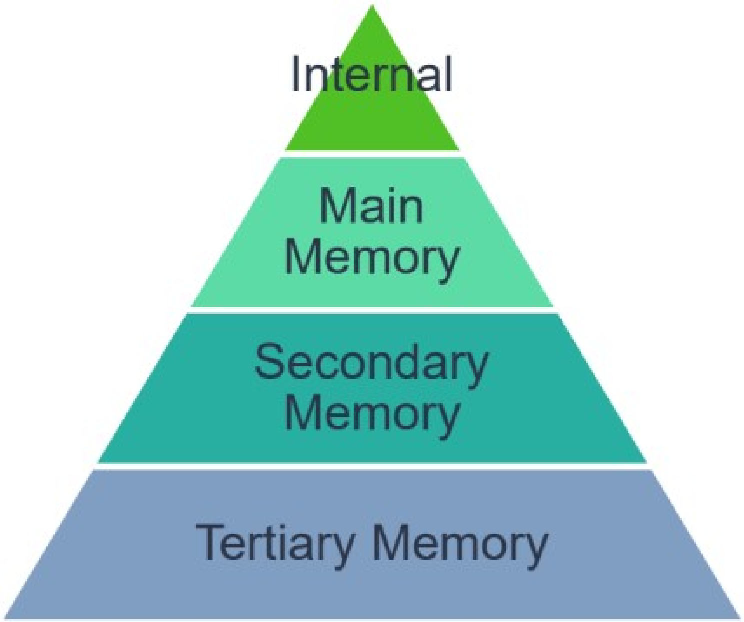
Let's Begin the Unlimited Learning

unacademy

Memory

Device to store content in computer system

Memory Hierarchy



Main Memory (Primary)

Main Memory

Used to store current running programs and their data

instructions

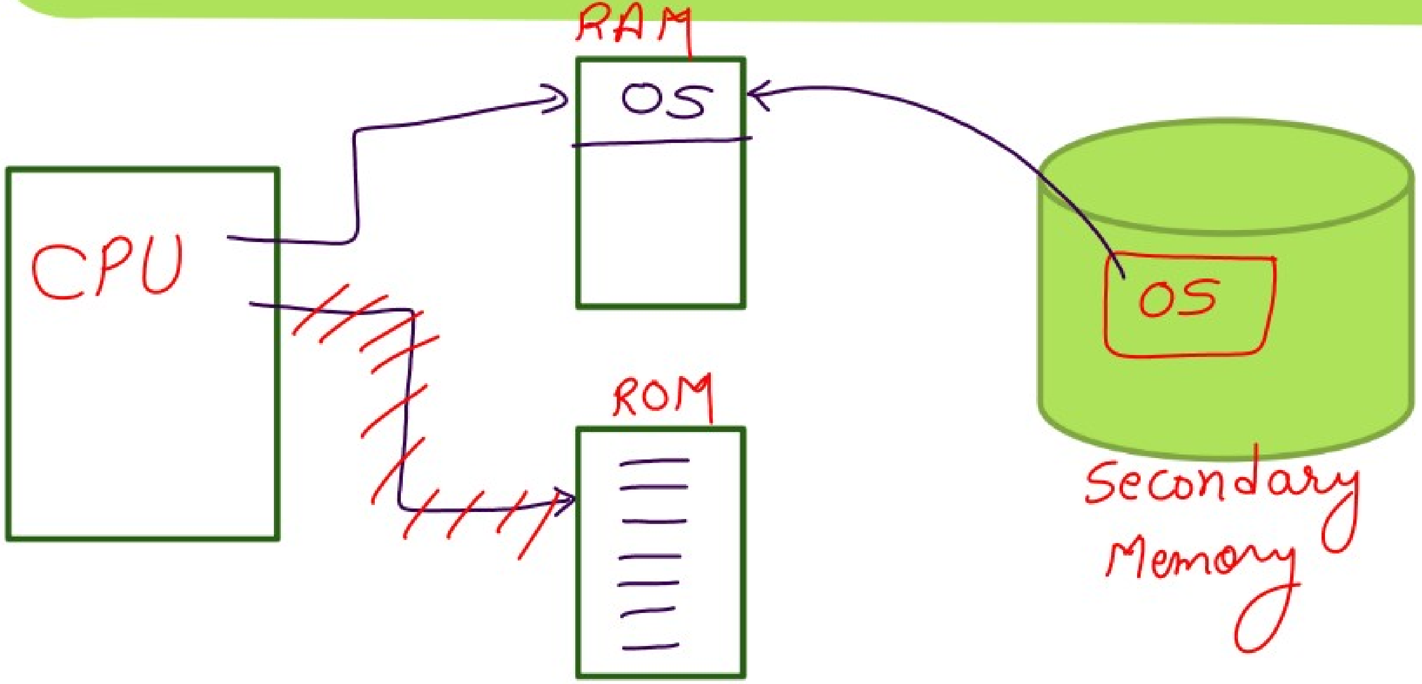
Main Memory

Used to store current running programs and their data

Types:

1. RAM (Random Access Memory) ⇒ volatile
2. ROM (Read only Memory) ⇒ non-volatile

ROM



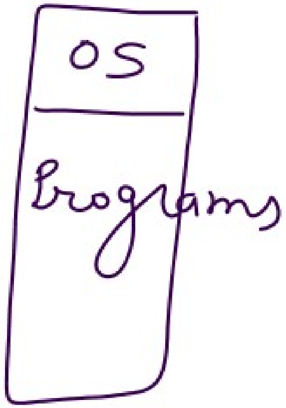
1. P.O.S.T.
2. Booting

ROM

A program, execution of which is responsible for booting is known as Bootstrap program.

ROM

RAM



unacademy

Types of RAM

Static (SRAM)	Dynamic (DRAM)
1. Made of → flip-flops	1. Capacitors ↳ storage in form of electric charges.
2. No refresh	2. Periodic Refresh is required to keep content.
3. Faster	3. slower
4. Expensive	4. Less Expensive
5. Used for cache	5. Used for m.m. implementation

Types of RAM

Static	Dynamic
1. Implemented using flip-flops	1. Implemented using capacitors
2. No refresh required	2. Periodic refresh is required
3. Faster Read/Write	3. Slow Read/Write
4. Used for Cache	4. Used for main memory
5. Low Idle power consumption	5. High Idle power consumption
6. High operational power consumption	6. Low operational power consumption

Table for Multiplication

Table of multiplication of two 2-bits unsigned integers.

a	b
2-bits	2-bits
00	00
01	01
10	10
11	11

Table

←4-bits→

$0*0$

$0*1$

$0*2$

⋮

$3*3$

Total no. of outputs to be stored $= 2^2 * 2^2$
$= 16$

Table ⇒ 16×4-bits

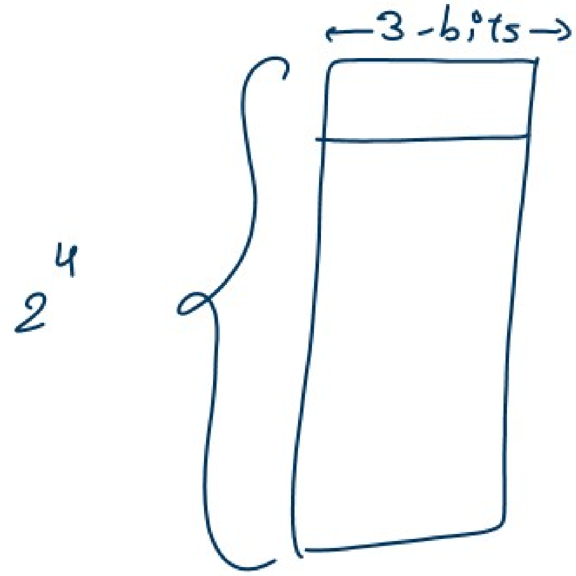
$2^{2n} \times 2n$-bits

Table for Multiplication

Input 1	Input 2	Multiplication Result Size	Multiplication Table Size	Addition Result	Addition Table Size
4-bits	4-bits	8-bits	$2^8 \times 8$-bits	5-bits	$2^8 \times 5$-bits
n-bits	n-bits	2n-bits	$2^{2n} \times 2n$-bits	(n+1) bits	$2^{2n} \times (n+1)$-bits

Table for ~~Multiplication~~ Addition

a	b
2-bits	2-bits
00	00
01	01
10	10
11	11



$2^4 \times 3$-bits

$2^{2n} \times (n+1)$ bits

unacademy

Question 1

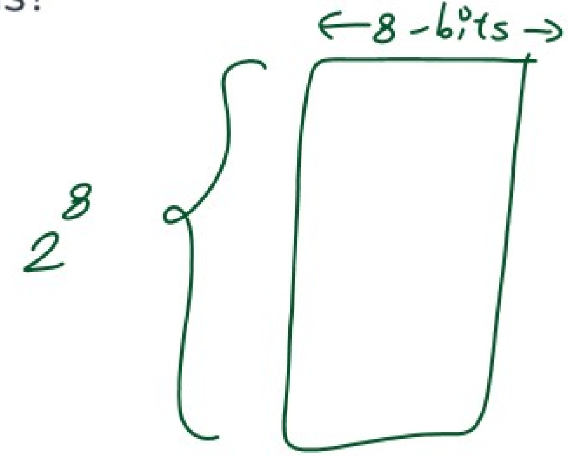
The amount of ROM needed to store the table for multiplication of two 4-bit unsigned integer is?

A) 64 bits

B) 128 bits

C) 1K bits

D) 2K bits



$2^8 \times 8$-bits

2^{11} bits

2K bits

D

unacademy

Question 2

A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is?

a) 256×16

b) 64K×8

c) 4K×16

d) 64K×16

$2^{16} \times 16$ - bits

64K × 16 bits

D